Entropy Collapse Has a Direction: Correcting the Visitation Channel in Token-Level Entropy Control for RLVR

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Abstract

Reinforcement learning with verifiable rewards (RLVR) reliably drives a policy's entropy toward zero, and a growing family of methods intervenes by re-weighting the policy loss with a per-token estimate of how much each token's update costs in entropy. We show that every such method estimates only half of the relevant quantity. The gradient of the *trajectory* entropy splits exactly into two terms: a *local* channel, which asks how the conditional distribution at a visited state flattens or sharpens, and a *visitation* channel, which asks how the probability of *reaching* that state changes. Existing token-level estimators capture the first and are structurally blind to the second. We show the blindness is worse than an omission: the local estimator is exactly zero at a uniform branch point, that zero is the correct first-order answer, and the visitation term's prefactor is simultaneously maximal there — the two channels are *anti-correlated*, so no refinement of the local term can recover the missing signal. We then derive an estimator for the visitation channel that costs one extra forward pass. The reward-to-go implied by the decomposition is the remaining conditional entropy H_{togo}, which we forecast with parallel multi-token-prediction heads; the sibling-prefix baseline that keeps the estimator unbiased is supplied by a tree-structured rollout in which several samples share a prefix and diverge at scheduled depths. The resulting advantage A_H sums to zero within a sibling set by construction, so the method cannot change the average amount of damping — it can only decide which of two siblings is damped. We give the full derivation, the four approximations it requires and what each discards, and a five-arm design that isolates the contribution of the forecast's *value* from that of the apparatus carrying it.

1 Introduction

Policy-gradient fine-tuning with verifiable rewards sharpens a language model's output distribution. Measured as the mean per-token entropy of the policy, this sharpening is fast, monotone, and, past a point, coincident with the end of useful learning. The standard reading is that the collapse is the problem, and the standard remedy is to spend part of the update budget slowing it down: an entropy bonus, a KL anchor, a clip on the tokens that lose the most entropy, or — in the most recent line of work — a per-token re-weighting of the policy loss by a first-order estimate of the entropy each token's update destroys.

We argue that the accounting behind that remedy is incomplete, and that the incompleteness is structural rather than a matter of estimator quality.

Write $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution the policy induces over whole responses to a prompt x. The autoregressive factorization turns it into an expectation of a sum of per-token conditional entropies, and the gradient of that expression splits, by the product rule, into exactly two terms. The first holds the visited states fixed and asks how the conditional distribution at each one changes: this is what a per-token entropy estimate measures. The second holds the per-state entropies fixed and asks how the probability of *visiting* each state changes. An update that concentrates mass on one continuation makes the states downstream of that continuation more likely and the states downstream of its abandoned siblings less likely; the diversity those siblings carried leaves the trajectory distribution even though no conditional distribution anywhere became more peaked.

Our first contribution is to show that the local channel is not merely blind to the second term but *anti-correlated* with it. The local estimator factors into the advantage, a saturation factor, and the deviation of the sampled token's surprisal from the mean surprisal — and that last factor has mean zero by the definition of entropy. At a branch point where several continuations are close to equiprobable it vanishes identically. The vanishing is not an estimator failure: a uniform distribution maximizes entropy, so the first-order change is genuinely zero and the damage is second order (Section 4.2). Meanwhile the visitation term's prefactor is $1 - \pi_\theta(y_t \mid s_t)$, which is maximal precisely at those flat states. The local channel therefore spends its budget where the conditional entropy is fragile — rare tokens sampled from confident distributions — and spends nothing where the trajectory entropy is most at risk.

Our second contribution is an estimator for the missing channel that requires no additional rollouts. Eliminating the non-causal cross terms turns the visitation term into a REINFORCE estimator whose reward-to-go is the sum of the *remaining* conditional entropies, H_{togo}; subtracting any function of the state leaves it unbiased. We estimate H_{togo} with parallel multi-token-prediction (MTP) heads read off the hidden state the policy already computes, and we obtain the state-only baseline from rollouts that share a prefix. Plain group-relative rollouts almost never share prefixes, so we schedule the sharing: a tree-structured sampler branches a single root at fixed depths, and the resulting sibling sets give the baseline an actual support (Section 9).

The resulting per-token score A_H is a deviation from a sibling mean and therefore sums to zero over each sibling set. This has a consequence we state up front because it determines what evidence can count: the method *cannot* raise the average weight at branch points, only spread the weights of siblings apart. "Total entropy went up" is therefore not a prediction of this method, and an experiment looking for it is testing a different hypothesis (Section 9.3).

We make the following contributions.

- An exact two-channel decomposition of the trajectory-entropy gradient, and an argument that the channel captured by current token-level entropy-control methods is anti-correlated with the channel they miss (Sections 4–4.2).
- A derivation of the missing channel into a per-token advantage A_H with no approximation up to the point of estimating H_{togo}, together with an explicit ledger of the approximations that estimation requires, the sign of each one's error, and the mechanism that absorbs it (Sections 5–10).
- A tree-structured rollout that supplies the sibling-prefix baseline, and a bound showing what fraction of positions can carry a non-zero A_H under *any* sampler (Section 9).
- A five-arm experimental design that separates the contribution of the forecast's value from the contribution of the apparatus that delivers it, by running the full pipeline twice with the forecast's value destroyed in two different ways (Section 11).

2 Related Work

Entropy collapse in RLVR. Entropy decay under policy-gradient fine-tuning is well documented, and a range of interventions add an explicit entropy term to the objective or constrain the update so that high-entropy positions are preserved. Recent analyses report that naive entropy regularization does not prevent the collapse and that continuing to train under it degrades accuracy (Anonymous, 2025d), which is consistent with our reading: the regularizer acts on the local channel, and the local channel is not where trajectory diversity is being lost.

Token-level entropy control. The method we build on re-weights the clipped surrogate loss by a per-token first-order estimate of the entropy change that token's update induces (Anonymous, 2025c). That work observes that earlier interventions are "indirect" and that their effect is therefore limited and can fail. Our decomposition supplies a mechanism for that observation: the estimate is a first-order estimate of the local channel only, and it is exactly zero at the states where the visitation channel is largest.

Which tokens carry the gradient. Anonymous (2025a) show that restricting policy-gradient updates to a small minority of high-entropy "forking" tokens matches full-gradient training. That result establishes that the identity of the tokens, not the aggregate entropy, is what matters, and it motivates a per-token treatment; it does not, however, distinguish the two channels, since forking tokens are identified by their local entropy. Our analysis predicts that a local criterion will select a systematically different set from the one the visitation channel would select.

Local stochasticity without global diversity. Anonymous (2025b) report that token-level entropy can remain high while the diversity of final answers falls. This is precisely the signature our decomposition predicts when the visitation channel is unmanaged: per-state entropies are preserved — the local channel is doing its job — while the trajectory distribution concentrates.

Multi-token prediction. We use Medusa-style parallel prediction heads (Cai et al., 2024) not for their intended purpose (speculative decoding) but as an entropy forecaster: we need the entropy of the distribution over a token k steps ahead, not a good sample of that token. Section 6 shows that this choice of estimator has a signed, monotone error which we characterize and correct.

3 Preliminaries

Notation. A prompt is x, a response is $y = y_{1:T}$, and the state before emitting token t is $s_t = (x, y_{<t})$. We write $H(\pi_\theta(\cdot \mid s_t))$ for the entropy of the conditional distribution at a single position (lowercase, per-token) and $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution over whole responses (uppercase, per-trajectory). The surprisal of a token a is $I_a := -\log \pi_\theta(a \mid s_t)$, so that $H = \mathbb{E}_{a\sim\pi_\theta}[I_a]$.

Group-relative policy optimization. We optimize with GRPO. For each prompt, n responses are sampled and scored, and each response receives the group-standardized advantage $A_i = (r_i - \mathrm{mean}_j r_j)/\mathrm{std}_j r_j$, broadcast to every token of that response. Replacing a learned value function with the group mean is what makes GRPO cheaper than PPO; it also means that a group whose members all receive the same reward contributes no gradient.

Token-level entropy control. The baseline intervention leaves the advantage untouched and instead multiplies each token's contribution to the clipped surrogate by a weight $\alpha_t \in [w_{\min}, w_{\max}]$ derived from a first-order estimate Ω_t of the entropy change that token's update induces. With $w_{\max} = 1$, the weight can only attenuate. Writing A_u for the advantage, $\pi_u := \pi_\theta(y_u \mid s_u)$ for the probability of the sampled token, and H_u for the local entropy, the estimator used in that line of work is, up to a positive constant,

$$\Omega_u \;=\; A_u\, \pi_u (1 - \pi_u) \left(\log \pi_u + H_u\right). \quad (1)$$

The policy loss becomes $\sum_u \alpha_u \cdot \text{PPO-clip}(A_u, r_u)$, where the weights are a monotone map of Ω over the batch.

4 Two Channels

4.1 The decomposition is exact

Applying the entropy chain rule to $\pi_\theta(y \mid x) = \prod_t \pi_\theta(y_t \mid s_t)$ gives an identity, not an approximation:

$$\mathcal{H}(\pi_\theta \mid x) = \mathbb{E}_{y\sim\pi_\theta}\Big[\sum_{t=1}^{T} H\big(\pi_\theta(\cdot \mid s_t)\big)\Big]. \quad (2)$$

The left-hand side is what we would like to protect; the right-hand side is what instrumentation can measure. The entire argument of this paper follows from the observation that the measure $\mathbb{E}_{y\sim\pi_\theta}$ depends on θ as well.

Set $f_\theta(y) := \sum_t H(\pi_\theta(\cdot \mid s_t(y)))$, so that $\mathcal{H} = \mathbb{E}_{y\sim\pi_\theta}[f_\theta(y)]$. Both the integrand and the measure depend on θ, so the product rule — with the score-function identity restoring the second term to an expectation — yields

$$\begin{aligned}\nabla_\theta \mathcal{H} = &\underbrace{\mathbb{E}\big[\nabla_\theta f_\theta(y)\big]}_{\text{(L) local}} \\ &+ \underbrace{\mathbb{E}\big[f_\theta(y)\, \nabla_\theta \log \pi_\theta(y \mid x)\big]}_{\text{(V) visitation}}.\end{aligned} \quad (3)$$

(L), the local channel, holds the visited states fixed and asks how the conditional distribution at each one changes. Equation (1) is a first-order estimate of exactly this.

(V), the visitation channel, holds the per-state entropies fixed and asks how the probability of reaching each state changes. An update that concentrates mass on one continuation makes the subtree below it more likely and the sub-trees below its siblings less likely. Nothing in Eq. (1) can see this: Ω reads the conditional distribution at a position and never reads the probability of arriving there.

The two terms can have opposite signs. A method that drives (L) to zero has not thereby bounded $\nabla_\theta \mathcal{H}$.

4.2 Why the local channel cannot be repaired

A natural objection is that the two channels might coincide in practice: branch points are positions where the policy is uncertain, so they are high-entropy positions, so Ω should be large there and the local term should protect them automatically. The premise is correct — in warm-up rollouts on our model, branch diversity correlates with local entropy at Spearman $\rho = 0.65$, and with the entropy-to-go at $\rho = 0.82$. The inference from it fails, in three steps.

Step 1: Ω reads a deviation, not a level. Rewriting Eq. (1) in terms of surprisal gives

$$\Omega \;=\; A\,\pi_a(1-\pi_a)\,\big(H - I_a\big), \tag{4}$$

and $H = \mathbb{E}_{a\sim\pi}[I_a]$ by definition. The factor $H - I_a$ is therefore a mean-zero fluctuation: what sets the magnitude of $|\Omega|$ is the *spread* of surprisal, not the *height* of the entropy.

Step 2: at a uniform branch point $\Omega \equiv 0$. If the distribution is uniform over k candidates then $I_a = \log k = H$ for every a, so Ω vanishes identically. The positions that are most branch-like — several continuations at comparable probability — are exactly where the local term contributes nothing.

Step 3: and that zero is the correct answer. This is the step that closes the argument. A uniform distribution maximizes entropy, so $\partial H/\partial z_a = 0$ for every logit z_a; Ω is not wrong, it is right. Perturbing one logit of a uniform distribution by δ and writing $S := e^{\delta} + k - 1$ gives the closed form

$$H(\delta) = \log S - \frac{\delta e^{\delta}}{S}, \qquad H''(0) = -\,\frac{k-1}{k^2}, \tag{5}$$

with $H'(\delta)$ of sign $-\operatorname{sign}(\delta)$: any push in any direction lowers the entropy, but the loss is entirely second order. For a 10-way uniform distribution and $\delta = 0.1$ the actual entropy loss is 4.7×10^{-4}, roughly two orders of magnitude below that of a peaked distribution under the same perturbation. *As far as local entropy is concerned, uniform branch points are genuinely safe*, and a local estimator that allocates them no protection is allocating correctly for its own objective. No refinement of the local term recovers the missing signal, because the missing signal is not local.

The two channels are anti-correlated. The visitation term's prefactor, derived in Section 7, is $1 - \pi_a$, which approaches 1 as the distribution flattens — and the probability that a group of n samples actually diverges at a position approaches 1 in the same limit. Table 1 evaluates both channels on a family of distributions that interpolates between confident and flat. $|\Omega|$ is dominated by positions where the model is fairly confident but a tail token was sampled; at those positions siblings diverge only about half the time. Where divergence is certain, the local term is identically zero and the visitation prefactor is maximal.

distribution	H	$P(\text{split})$	$\mathbb{E}\lvert\Omega\rvert/\lvert A\rvert$	$\mathbb{E}[1-\pi_a]$
q=.99	0.10	0.08	0.092	0.020
q=.9, tail	1.25	0.57	**1.129**	0.190
q=.6, m=3	1.11	0.98	0.457	0.587
unif. k=10	2.30	**1.00**	**0.000**	**0.900**
unif. k=100	4.61	**1.00**	**0.000**	**0.990**

Table 1: The local channel and the visitation channel are ordered oppositely. Distribution family: top token has mass q, the remaining $1-q$ is spread uniformly over m tokens. $P(\text{split})$ is the probability that n=8 samples do not all agree. $\mathbb{E}|\Omega|$ is taken over the token actually sampled at that position.

The statement we can therefore make is stronger than "the local term is incomplete": *the local term is anti-correlated with the term it omits*, and the omission cannot be fixed within the local channel.

5 Estimating the Visitation Channel

5.1 Causal elimination gives a reward-to-go

Substituting $\nabla_\theta \log \pi_\theta(y \mid x) = \sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u)$ into (V) produces a double sum over (t, u). For $u \geq t$ the entropy $H(\pi_\theta(\cdot \mid s_t))$ is a function of $y_{<t}$, hence measurable with respect to $y_{<u}$, and the inner conditional expectation vanishes:

$$\mathbb{E}\big[\nabla_\theta \log \pi_\theta(y_u \mid s_u) \mid y_{<u}\big] = \nabla_\theta \sum_a \pi_\theta(a \mid s_u) = 0.$$

Only $u < t$ survives, and the surviving inner sum is the entropy still to come:

$$(\mathrm{V}) = \mathbb{E}\Big[\sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u)\, H_{\text{togo}}(s_u \oplus y_u)\Big], \tag{6}$$

with $H_{\text{togo}}(s_u \oplus y_u) := \sum_{t>u} H(\pi_\theta(\cdot \mid s_t))$.

We stress that H_{togo} was not posited. Equation (6) is a REINFORCE estimator whose reward is "the conditional entropy remaining after this token", and H_{togo} is that reward-to-go.

5.2 The sibling-prefix baseline

Any function $b(s_u)$ of the state alone leaves Eq. (6) unbiased, by the same argument. Subtracting one gives the per-token advantage

$$A_H(s_u, y_u) := H_{\text{togo}}(s_u \oplus y_u) - \bar{H}_{\text{togo}}(s_u), \tag{7}$$

and up to this point *no approximation has been made*.

The baseline we use is the mean of H_{togo} over the rollouts in the same prompt group whose prefixes $y_{<u}$ agree exactly. Those rollouts share the state s_u by definition, so their mean is a function of s_u alone and the baseline is admissible. The condition is agreement on $y_{<u}$ and not on $y_{\le u}$: the rollouts must diverge *at* u for the difference to be a branch score.

Two properties follow immediately and matter later. First, the implementation includes a rollout in its own sibling set, so b depends weakly on y_u and the estimator is not strictly leave-one-out; the resulting bias is $O(1/m)$ in the number m of surviving siblings, the same property GRPO's group-mean advantage has. Second, when $m = 1$ the baseline equals the rollout's own value and $A_H \equiv 0$, so a position with no siblings is silent rather than noisy.

6 Four Approximations

Equation (7) is exact but not computable: H_{togo} sums conditional entropies at states the update has not visited. Three approximations make it computable and a fourth bounds its influence.

6.1 A1: horizon truncation and discounting

We truncate the sum to κ steps and discount:

$$H_{\text{togo}}(s_u \oplus y_u) \approx \sum_{k=1}^{\kappa} \gamma_H^k \, H\big(\pi(\cdot \mid s_{u+k})\big). \quad (8)$$

The discount serves two purposes: distant terms carry more forecast error (A2) and therefore deserve less trust, and the sum stays bounded independently of κ.

What this discards — the tail $\sum_{k>\kappa}$ and a fraction of the retained terms — is largely harmless for a reason specific to our use. Equation (7) consumes only the *difference* between siblings. Any bias common to rollouts sharing s_u cancels in that difference, and truncation bias is common to first order.

6.2 A2: forecasting the unrealized future

Even Eq. (8) requires the future states s_{u+k}. At update time they do not exist: rollouts came from π_{old}, each state has exactly one sampled continuation, and re-rolling is prohibitive. We replace the realized future with a parallel forecast from the current hidden state h_u:

$$H\big(\pi(\cdot \mid s_{u+k})\big) \;\longrightarrow\; H\big(p_{\text{MTP}}^{(k)}(\cdot \mid h_u)\big). \quad (9)$$

Head k is a two-layer residual MLP applied to h_u, followed by the policy's own unembedding; the heads own no vocabulary parameters. Their last layer is zero-initialized, so before any training every head reproduces the policy's own next-token distribution and the forecast starts from a sane value. In the RL path the logits are materialized in chunks and reduced to an entropy immediately, so the $[\kappa, B, T, V]$ tensor never exists.

The error of this substitution has a known sign. Conditioning on the intervening tokens is exactly what the forecast omits. Fixing S_u, the k-th term of Eq. (8) has expectation $H(Y_{u+k} \mid Y_{u+1:u+k-1}, S_u)$, whereas the MTP head models the *marginal* $p(Y_{u+k} \mid S_u)$. Since conditioning reduces entropy,

$$\begin{aligned} H(Y_{u+k} \mid S_u) - \mathbb{E}\big[H(\pi(\cdot \mid S_{u+k-1}))\big] \\ = I\big(Y_{u+k};\, Y_{u+1:u+k-1} \mid S_u\big) \\ \ge 0. \end{aligned} \quad (10)$$

The forecast *always over-estimates*, and the excess is precisely the mutual information with the intervening tokens, which grows monotonically with k.

Equation (10) is not merely a bound: it is the reason the marginal is the quantity we want. It is the fan-out of the continuation — how many ways the next k tokens can unfold — rather than the residual uncertainty along the one path that happened to be sampled.

6.3 A3: per-head affine calibration

Equation (10) predicts a per-head, monotone inflation, so we absorb it with a per-head affine correction fitted by least squares against oracle entropies on warm-up rollouts:

$$\hat{H}_k = a_k \, H\big(p_{\text{MTP}}^{(k)}(\cdot \mid h_u)\big) + b_k. \quad (11)$$

The fitted coefficients confirm the prediction. On our model the first head has $a_1 = 1.03$ — a one-step forecast has no intervening tokens, so the mutual information in Eq. (10) is zero and the approximation is exact — while a_k drops to 0.10–0.19 for $k \ge 2$. Calibration switches the unreliable heads off on its own.

Combining A1–A3 gives the forecast that is actually computed:

$$\hat{H}_{\text{togo}}(s_u \oplus y_u) = \max\Big(0,\ \sum_{k=1}^{\kappa} \gamma_H^k \big(a_k H_k(h_u) + b_k\big)\Big), \tag{12}$$

the clamp encoding the fact that an entropy cannot be negative. With $\kappa = 2$ and $\gamma_H = 0.7$ this instantiates to $\hat{H}_{\text{togo}} = 0.72\, H_1 + 0.06\, H_2 + 0.11$: the second head contributes about 8%, so the effective horizon is close to one step. We regard this as a consequence of Eq. (10) rather than a tuning artifact, and note it as a limitation — a forecaster whose error did not grow with k would leave a longer horizon usable.

Alignment. One detail decides whether the signal is a branch score at all. The hidden state at position u is conditioned on s_u and predicts y_u; it does not know y_u. The quantity we need, $H_{\text{togo}}(s_u \oplus y_u)$, must therefore be read from position $u+1$ and shifted back by one. Omitting the shift measures the value of the state *before* the branch rather than the value of the branch taken.

7 From Gradient to Token Weights

Equation (7) is a term to add to a gradient, but methods in this family do not add gradients — they re-weight a loss. What is needed is a first-order prediction of how much one update, acting through token u, changes the trajectory entropy.

How far one step moves a logit. With $w_u = \text{clip}(r_u, 1-\epsilon_{\text{lo}}, 1+\epsilon_{\text{hi}})A_u$ the coefficient the policy loss actually applies, a single step moves only the sampled token's logit, $\Delta z_{u,a} = \eta\, w_u\, \delta_{a,y_u} + O(\eta^2)$. The softmax Jacobian $\partial \log \pi_a / \partial z_a = 1 - \pi_a$ then gives

$$\widehat{\Delta \log \pi_u} = \eta\, w_u\, (1 - \pi_u), \tag{13}$$

which vanishes as $\pi_u \to 1$: a token that already holds the mass cannot be given more. This is the prefactor whose anti-correlation with $|\Omega|$ Table 1 records.

The visitation-channel prediction. Because $\nabla_\theta \log \pi_u \cdot \Delta\theta = \Delta \log \pi_u$, substituting into Eq. (7) gives the per-token quantity

$$\text{visit}_u := \widehat{\Delta \log \pi_u} \cdot \text{clip}\big(A_H(s_u, y_u), -c, +c\big). \tag{14}$$

A4 is the clip, and it is applied to A_H *before* the product rather than after, so that one token's influence is bounded by $|\eta w_u| \cdot c$. Clipping the product instead would let forecast error be amplified without bound at tokens with large w_u.

Sign quadrants. Equation (14) defines the collapse channel precisely. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H > 0$, mass moves toward a branch richer than its siblings and trajectory entropy rises. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H < 0$, mass moves into a dead end and the diversity carried by the abandoned siblings is destroyed — this is the collapse path that (L) is structurally unable to see. Negative $\widehat{\Delta \log \pi_u}$ reverses both readings.

8 Combining the Channels

The two channels enter the same weight:

$$\tilde{\Omega}_u = \text{norm}(\Omega_u) + \lambda \cdot \text{norm}(\text{visit}_u), \tag{15}$$

after which $\tilde{\Omega}$ is passed to the unmodified min–max mapping of the base method, producing $\alpha_u \in [w_{\min}, w_{\max}]$. We emphasize that the hook is inserted where Ω is finalized and that nothing downstream is changed; when $\lambda = 0$ the pipeline is bitwise identical to the base method, which we enforce with a regression test.

Three choices in Eq. (15) deserve comment. Normalization is required even though both terms carry units of nats, because Ω contains a factor $1/\pi_{\text{old}}$ and is heavy-tailed. Because normalization removes scale, the step size η in Eq. (13) only ever appears multiplied by λ; we fix $\eta = 1$ and sweep λ alone. Finally, norm is an RMS rescaling rather than a z-score, since subtracting a mean would break the $\lambda = 0$ equivalence.

9 Tree-Structured Rollouts

9.1 Why plain rollouts cannot support A_H

$\hat{H}_{\text{togo}}$ is a deterministic function of the causal prefix, so two rollouts sharing $y_{\le u}$ produce identical forecasts. Combined with Eq. (7) this gives a sharp characterization:

$$A_H[i,u] \neq 0 \iff \begin{array}{l}\text{siblings share } y_{<u} \\ \text{but diverge at } y_u,\end{array} \tag{16}$$

that is, u is a branch point of the group. Under independent sampling from a sharpened policy, group members diverge at their first token and never share a prefix again, so $m = 1$ almost everywhere and $A_H \equiv 0$: the visitation channel is not merely weak, it is identically absent.

We therefore schedule the sharing. A tree sampler expands a single root and cuts it at fixed depths

$d_1 < d_2 < d_3$ with branching factor 2, yielding $n = 8$ leaves whose sibling structure is known in advance. The cuts create the only positions at which Eq. (16) can fire.

9.2 A ceiling on the support

It is tempting to treat the fraction of positions with $A_H \neq 0$ as a quantity to be maximized. It is not, and the reason is combinatorial rather than empirical. Two quantities behave very differently under the tree:

- the fraction of positions where *some* other rollout shares $y_{<u}$ is moved by the cuts, and equals d_L/T by construction;
- the fraction where $A_H \neq 0$ is *not*, because by Eq. (16) it counts branch points, and n sequences form a trie with at most $n-1$ internal branching nodes.

Since the one-position alignment shift widens each branch node into two columns, the fraction of positions carrying a non-zero A_H is bounded by $2(n-1)/T$ for *any* sampler whatsoever. With $n = 8$ and responses of $\sim 10^3$ tokens this ceiling is on the order of a percent. The tree's role is to make the branch points exist and to place them at controlled depths, not to make them numerous.

9.3 The correction cannot move the mean

A_H is a deviation from a sibling mean, so on each sibling set

$$\sum_{j \in \mathrm{sib}(s_u)} A_H\big(s_u, y_u^{(j)}\big) = 0. \qquad (17)$$

The correction in Eq. (15) therefore cannot change the average weight assigned to branch tokens. It can only spread siblings apart. This is not a defect to be engineered away: an unbiased estimator requires a baseline, and a baseline forces the mean to zero.

The consequence for evaluation is direct, and we state it before presenting any experiment. "Mean entropy at branch points increased" is *not* a prediction of this method and cannot serve as evidence that it works; the corresponding prediction is that the *variance across siblings* widens. A method designed to raise the mean is a different and weaker hypothesis — it does not need a forecast at all, only a support — and we include it below as a control arm.

10 The Approximation Ledger

Table 2 collects every departure from Eq. (3).

The strongest assumptions are A2 and B2. A2 has a known sign and a known monotonicity (Eq. 10), which is what makes the affine correction in A3 an appropriate remedy rather than a curve fit. B2 — that updating one token does not change the conditional distribution at other positions — is false for a network with shared parameters and has no corrective mechanism. It is not a risk this work introduces, since the base method already relies on it, but adding a second channel adds a second copy of its error.

11 Experimental Setup

11.1 Model, data, and optimization

All runs fine-tune the same base model on the same data with the same optimizer settings, and differ only in the intervention. We use a 1.5B-parameter mathematics-specialized model, train on a 17k-problem competition-mathematics set, and validate on AIME 2024 (30 problems) with 32 samples per problem. With a training batch of 512 prompts, 110 optimization steps correspond to about 3.2 epochs.

Optimization uses GRPO with $n = 8$ responses per prompt, a constant learning rate of 1e$-$6 with no warm-up (so that the schedule is independent of the declared total step count), a maximum response length of 3072 tokens, no entropy bonus, and no KL loss. Rollouts are sampled at temperature 1.0 with no nucleus truncation; validation uses temperature 1.0 and top-p = 0.7. Every arm uses the same data seed and the same initial checkpoint, which we verify by requiring the step-0 validation scores to agree.

11.2 Method hyperparameters

The forecaster uses $\kappa = 2$ heads with discount $\gamma_H = 0.7$ and the per-head calibration of Eq. (11). The combination uses $\lambda = 0.25$, RMS normalization, clip $c = 1.0$, the sibling-prefix baseline, and the base method's min–max mapping into $[w_{\min}, w_{\max}] = [0.7, 1.0]$ — so the weight can only attenuate, never amplify. The tree sampler uses a single root, cuts at depths $\{64, 192, 384\}$, and branching factor 2 at each cut, giving $n = 8$ leaves.

11.3 Five arms

The design isolates one question: does the *value* of the forecast matter, or only the apparatus that car-

	approximation	what it discards	sign / bound	mitigation
—	Eqs. (2)–(7)	nothing — identities	—	—
A1	horizon κ, discount γ_H	the tail $\sum_{k>\kappa}$, and a fraction of the retained terms	under-estimate	the common component cancels in the sibling difference
A2	realized future $\to$ MTP marginal	mutual information with intervening tokens, Eq. (10)	**over-estimate, monotone in k**	A3
A3	per-head affine calibration	non-affine residual of A2	least-squares residual	oracle-forecast control arm
A4	clip A_H to $\pm c$	magnitude of extreme branch scores	bounded by c	saturation rate is logged
B1	first order in η	$O(\eta^2)$	—	small learning rate
B2	only the sampled logit moves	cross-position effects of weight sharing	—	*inherited from the base method*
B3	rollouts come from π_{old}	mismatch with the π_θ expectation	—	importance ratio restores r
B4	self included in sibling set	$O(1/m)$ relative to leave-one-out	—	same property as the GRPO advantage

Table 2: Every approximation between the exact decomposition and the implemented weight. A1–A4 are introduced by this work; B1–B4 are shared with the token-level entropy-control method we extend.

arm	λ	rollout	signal in the weight
GRPO	—	plain	none
STEER	0	plain	local Ω only
uniform	.25	tree	support only
permuted	.25	tree	A_H shuffled
STEER-F	.25	tree	A_H as derived

Table 3: The five arms. UNIFORM and PERMUTED run the full pipeline with the forecast's value destroyed in two different ways, so any gain they show is attributable to the apparatus rather than to the visitation signal.

ries it? A comparison between the base method and the full method changes two things at once (λ and the sampler), so it cannot answer this. We therefore add two arms that keep the entire apparatus — the tree, the forecaster, the support, the weight mapping — and destroy only the information in A_H.

UNIFORM applies the same attenuation to every position in the support, discarding A_H's value but keeping its support — this is the implementation of the "raise the mean" hypothesis that Section 9.3 distinguishes from ours. PERMUTED keeps the multiset of A_H values within each sibling set and shuffles the assignment, so the support, the magnitudes, and Eq. (17) all survive while the pairing between a branch and its score is destroyed. Together they bracket the claim: if the visitation signal is doing the work, both controls should behave like the no-forecast arms.

11.4 Metrics

We report $\mathrm{acc@32}$ (mean accuracy over 32 samples) and $\mathrm{maj@32}$ (accuracy of the majority vote over the same 32 samples), and their difference

$$\mathrm{uplift} := \mathrm{maj@32} - \mathrm{acc@32},$$

which measures how much better a set of 32 samples is than its average member — that is, whether correct answers concentrate while incorrect ones scatter. We use uplift rather than $\mathrm{pass@32}$ because the latter does not separate the arms.

Because single validation points are noisy on a 30-problem set, the primary statistic is the mean over the converged window, steps 40–110 (8 validation points per arm). We quote differences in units of the across-problem standard error, $\mathrm{SE} = \mathrm{std@32}/\sqrt{30} \approx 0.027$, which is the correct scale for generalization to new problems. We separately note that the pure re-sampling noise of the protocol — estimated from two validations of identical weights — is about 0.004 in accuracy, an order of magnitude smaller; the two scales answer different questions and we are explicit about which is in use.

One instrumentation caveat propagates into every entropy number we report. The logged `actor/entropy` is aggregated with a token-mean, so it is $\mathcal{H}$ divided by the mean response length, not $\mathcal{H}$. Since response lengths differ by under 5% across arms, we report the raw quantity and, where the distinction could matter, the length-corrected product as well.

12 Results

[TODO: This section is deliberately left as scaffolding. The training logs for four of the five arms are available; the GRPO arm's log and the multi-benchmark evaluation are pending,

and the numbers will be filled in once both are verified from source.]

12.1 Main comparison

[TODO: Table: per-arm plateau means over steps 40–110 — acc@32, maj@32, uplift, entropy, mean response length, length-corrected entropy. Five rows, one per arm of Table 3.]

12.2 Does the forecast's value matter?

[TODO: Table of pairwise contrasts in SE units: STEER-F versus each of GRPO, STEER, uniform, permuted, and the three null contrasts among the control arms. The prediction of Section 11 is that the first group separates and the second does not.]

12.3 Decomposing the gain

[TODO: Split the maj@32 gain into the part attributable to per-sample accuracy and the part attributable to uplift, and split the total gain into the apparatus component (permuted minus base) and the information component (STEER-F minus permuted).]

12.4 Entropy is not the channel

[TODO: Table contrasting entropy gaps against accuracy gaps across arms, testing whether arms with matched aggregate entropy differ in accuracy. Section 9.3 predicts that they can, and that aggregate entropy therefore does not order the arms.]

12.5 Mechanism diagnostics

[TODO: Support fraction, mean sibling-set size, and branch-point fraction per step, against the $2(n-1)/T$ ceiling of Section 9; the two-term balance $\lambda\|\text{visit}\|/\|\Omega\|$; and the A_H clip saturation rate.]

12.6 Cost

[TODO: Seconds per step per arm and the resulting wall-clock overhead, attributed to the MTP forward pass.]

13 Conclusion

The gradient of trajectory entropy has two terms, and the token-level entropy controls now in use estimate one of them. We showed that the omission is structural: the estimated term vanishes exactly where the omitted term is largest, and it vanishes for a correct reason, so the gap cannot be closed by improving the estimator. We then showed that the omitted term is a REINFORCE estimator whose reward-to-go is the remaining conditional entropy, that a parallel forecaster estimates it for the price of one forward pass with an error whose sign and monotonicity are known in closed form, and that a tree-structured rollout supplies the baseline it requires. The resulting correction is mean-preserving by construction: it does not slow entropy collapse, it redirects it.

Limitations

A single validation benchmark. Every claim rests on AIME 2024, which has 30 problems. At SE ≈ 0.027, differences of interest are a fraction of one problem, and a multi-benchmark evaluation is required before the direction of any gap can be called robust. We report the standard error explicitly rather than significance stars for this reason.

A single seed. All arms use one data seed. The standard error we quote is an across-problem quantity and does not bound run-to-run variation, which for these runs is uncontrolled because rollout sampling is not seeded across processes. A re-run of the same configuration is a second draw, not a restoration.

The effective horizon is one step. The calibration in Eq. (11) suppresses heads beyond the first, so $\hat{H}_{\text{togo}}$ is dominated by a single step ahead. This is the predicted behavior of Eq. (10) rather than a bug, but it means the "future" in the visitation term is a short future, and a forecaster whose error did not grow with k would test the idea more sharply. A sequential forecaster is the obvious comparison and we have not run it.

The support is small by construction. Section 9 bounds the fraction of positions carrying a non-zero A_H at $2(n-1)/T$, on the order of a percent for our settings. Whether the effect scales with n, or with a branching schedule chosen adaptively rather than at fixed depths, is untested.

An assumption with no remedy. Approximation B2 — that a token's update does not change the conditional distribution at other positions — is false in a network with shared parameters. It is inherited from the base method rather than introduced here, but our construction adds a second estimator that relies on it.

One prediction of the analysis is unmeasured. Section 4.2 argues that the two channels are anti-correlated. The prefactor half of that claim is analytic and the A_H half is inferred from a correlation with branch diversity; we have not measured their *product* at real sibling-divergence positions in a live rollout. Until that measurement exists, Table 1 is a prediction rather than an observation.

Scale. All experiments use a single 1.5B model. The relative size of the two channels plausibly depends on how sharp the base policy already is, and we have no evidence about larger models.

Ethics Statement

This work studies an optimization-time regularizer for reinforcement learning on mathematics problems with automatically verifiable answers. It introduces no new data collection, no human subjects, and no new model release. The intervention acts on which of several equally-scored continuations receives a slightly smaller gradient weight; it does not alter the reward, and so does not change what behavior is being optimized for. The training data is a public competition-mathematics set and the evaluation set is a public examination.

The main foreseeable risk is indirect. Methods that preserve diversity during RL fine-tuning are not specific to mathematics, and the same mechanism would apply to a reward model encoding any objective, including a harmful one; we note that our method inherits, and does not mitigate, whatever the reward function encodes. The computational cost of the method (a forward pass through a set of small prediction heads at every optimization step) is a real environmental cost, which we quantify in Section 12.6 so that it can be weighed against the benefit.

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A Closed Forms for Section 4.2

[TODO: Derivation of Eq. (5) and the numerical table of first-order prediction versus actual entropy change for peaked and uniform distributions.]

B Implementation Notes

[TODO: Tensor layout of the sibling mask, the one-position alignment shift, the chunked entropy reduction that avoids materializing the head logits, and the $\lambda = 0$ bitwise-equivalence test.]